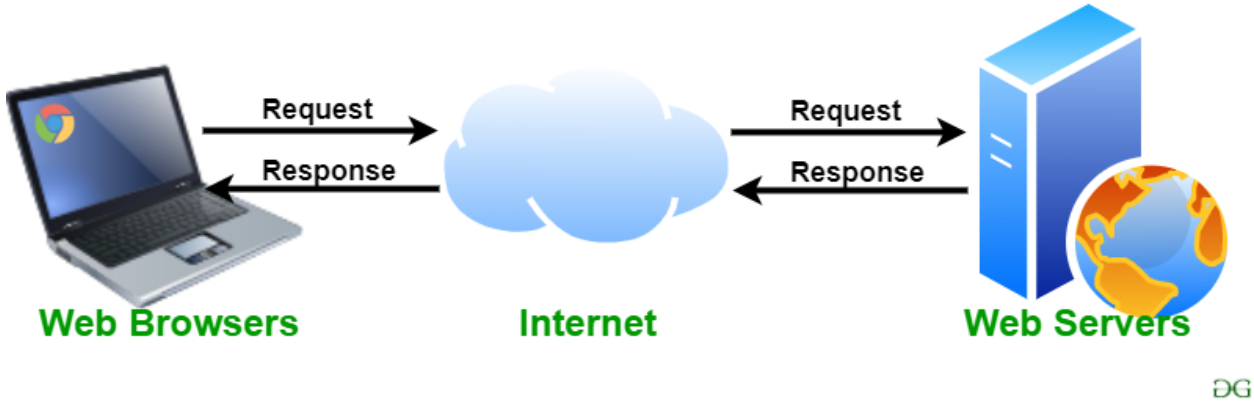
Frontend

Frontend is the part of a website that users see and interact with. It is built using HTML, CSS, and JavaScript.

Frontend = Structure + Style + Behavior
HTML = Structure
CSS = Design
JS = Logic

Working principle of web



- Server
  - server is the place where all the resources are present.
  - it accepts all the upcoming request.
- Web browser
  - Web browser is a application through which we can interact with different websites. Ex : chrome, firefox, edge
- Request
  - Data exchange from client (browser) to server.
  - Ways of sending the request to server

    1. by typing the URL in the browser

    2. by clicking any link or hyperlink

    3. by submitting the information

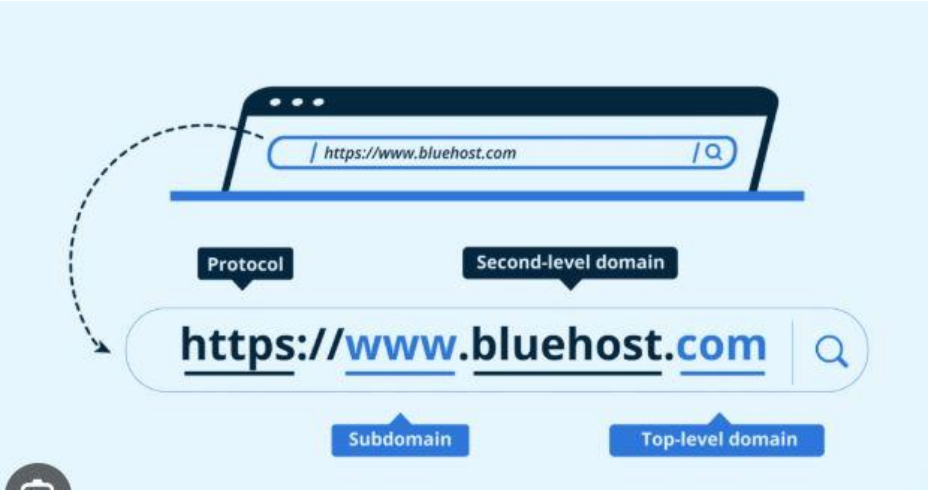
- Response
  - Data exchange from sever to client (browser)
- Website
  - Website is group of multiple webpages which are grouped together.
- Webpages

- Webpage is a document which is created by using any markup language and it can be displayed in browser
- Trough browser we can interact with any webpage

- www
  - it is an acronym for world wide web
  - a group of multiple computer which are grouped together to share information or communication via internet
- URL
  - URL is uniform resource locator
  - URL helps us to access webserver
  - Indicate location of web resource

DNS :

- Domain name service , internet phonebook which converts domain name or URL into IP address.



- Protocol
  - A set for rules which we use for communication
  - Ex : HTTP, FTP , HTTPS

- Search Engine
  - A web service that helps us to find other web pages.

HTML

Hypertext Markup Language is used to provide structure to webpage.

Father of HTML is tim berner's lee

HTML5 is current version

Hypertext : HyperText simply means Text within Text. A text has a link within it, is a hypertext. Whenever you click on a link which brings you to a new webpage, you have clicked on a hypertext. HyperText is a way to link two or more web pages (HTML documents) with each other.

Markup language : markup language is a computer language which helps us to create structure and layout for any text document. Markup language makes text more interactive and dynamic. It can turn text into images, tables, links, etc.

HTML5 Boilerplate is the **basic structure/template of every HTML5 document**.

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Document</title>
</head>
<body>
   // visible content
</body>
</html>
```

`<!DOCTYPE html>` : tells the browser that this is an HTML5 document.

Ensures browser uses **standard mode**, not old compatibility mode.

`<html lang="en">` : This is the **root element** of the entire page, Everything must be inside `<html>`.

`lang="en"` Specifies the language of the document that is english

`<head>` : The `<head>` contains **metadata** (information about the webpage), Users cannot see this content directly.

<meta charset="UTF-8"> : Defines character encoding, UTF-8 supports almost all characters and symbols.

<meta name="viewport" content="width=device-width, initial-scale=1.0"> :

- Makes your website responsive on mobile devices.
- `width=device-width` → page width matches device width
- `initial-scale=1.0` → initial zoom level

<title>My Website</title> : This appears: On browser tab

`<body>` : The `<body>` contains everything the user sees:

- ## Tags
  - Tag is a keyword in HTML which is enclosed inside angular brackets

paired tags / container tags : opening and closing tags ex, <h1></h1>, <p>

unpaired tags/ empty tags/ self closing tags/ void tags : contains only opening tags
 <hr>

- ## Elements

Opening and closing tags along with the content

Ex : <h1 > element </h1>

Attributes :

- Attributes are properties tags.
- All HTML elements can have attributes
- We can give attributes in opening tag
- Attributes provide additional information about elements
- Attributes usually come in name/value pairs like: **name="value"**

Comments

- Syntax : <!-- Write your comments here -->

<u>Absolute Path & Relative Path</u>

A **relative path** tells the browser the location of a file **relative to the current HTML file location**.

It depends on project folder structure.

Ex : <img src="image.jpg" alt="Image">

An **absolute path** gives the complete address of the file.

It does not depend on folder structure.

Ex : <img src="https://www.example.com/images/photo.jpg" alt="Image">

- **image tag**

  - The <img> tag is used to embed an image in an HTML document.

  - It is a **self-closing tag** (void element).

  Syntax : <img src="image.jpg" alt="description">

- **list tag** : HTML lists allow web developers to group a set of related items in lists.

1. unordered list : An unordered list starts with the <ul> tag. Each list item starts with the <li> tag.

The list items will be marked with bullets (small black circles) by default (types : disc, circle, square)

2. ordered list : An ordered list starts with the <ol> tag. Each list item starts with the <li> tag.

The list items will be marked with numbers by default (types : 1, A, a, I, i)

3. description list : A description list is a list of terms, with a description of each term.

The <dl> tag defines the description list, the <dt> tag defines the term (name), and the <dd> tag describes each term

- **anchor tag**
  - The <a> tag defines a hyperlink used to link one page to another.

- **pre tag** - used to display text as it is on screen

Iframe :

An HTML iframe is used to display a web page within a web page.

An inline frame is used to embed another document within the current HTML document.

The HTML <iframe> tag specifies an inline frame.

Syntax : `<iframe src="url" title="description"></iframe>`

Every HTML element has a default display value, depending on what type of element it is.

Table tag :

"Whenever data is arranged in rows and columns, we use a table."

Can include text, images, links, and other elements.

- Cells can include text, images, lists, links, or even another table (nested table).

`<table>` - Creates the table
`<tr>` - Table row
`<td>` - Table data (cell)

`<th>` - Makes text **bold**, Centers text, Used for headings

`<thead>` - Table header
`<tbody>` - Table body
`<tfoot>` - Table footer

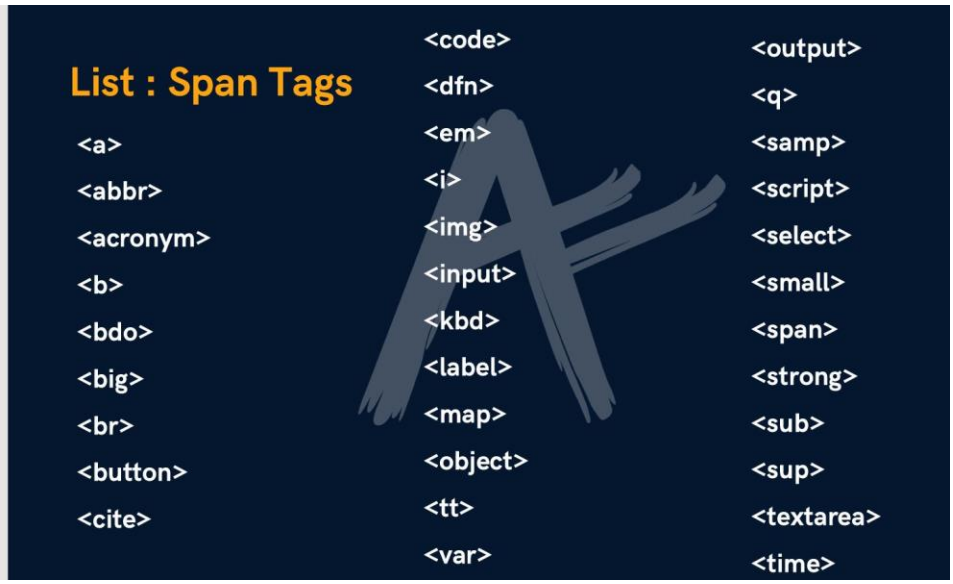
Attributes
Border - adds border
Cellpadding - space inside cell
Cellspacing - space between cells
Colspan - Combines columns.
Rowspan - Combines rows.

Block elements:

- **Start on a** new line
- **Take** full width of the parent container
- **Can contain** inline and other block elements

Inline elements:

- Do **not start on new line**
- Take **only required width**
- Cannot contain block elements



Inline-Block

Inline-block elements behave like inline elements but allow width and height.

Semantic elements

Semantic elements are HTML tags that clearly describe the meaning or purpose of the content they contain.

They make the code more readable and meaningful.

Tag	Purpose
`<header>`	Top section of webpage
`<nav>`	Navigation links
`<section>`	Section of content
`<article>`	Independent content
`<aside>`	Sidebar content
`<footer>`	Bottom section
`<main>`	Main content area
`<figure>`	Image with caption
`<figcaption>`	Image description

Non-semantic

Non-semantic elements do not describe the meaning of the content.

They are just containers for styling or grouping elements.

Form tag

HTML Form is used to **collect user input and send it to the server for processing**.

Input types :

- <input type="button">
- <input type="checkbox">
- <input type="color">
- <input type="date">
- <input type="datetime-local">
- <input type="email">
- <input type="file">
- <input type="hidden">
- <input type="image">
- <input type="month">
- <input type="number">
- <input type="password">
- <input type="radio">
- <input type="range">
- <input type="reset">
- <input type="search">

- <input type="submit">
- <input type="tel">
- <input type="text">
- <input type="time">
- <input type="url">
- <input type="week">

Form elements :

- <input>
- <label>
- <select>
- <textarea>
- <button>
- <fieldset>
- <legend>
- <datalist>
- <option>

Form Attribute :

action - The action attribute defines the action to be performed when the form is submitted, the form data is sent to a file on the server when the user clicks on the submit button.

target - The target attribute specifies where to display the response that is received after submitting the form.

method - The method attribute specifies the HTTP method to be used when submitting the form data.

autocomplete - The autocomplete attribute specifies whether a form should have autocomplete on or off, When autocomplete is on, the browser automatically complete values based on values that the user has entered before.

novalidate - The novalidate attribute is a boolean attribute, When present, it specifies that the form-data (input) should not be validated when submitted.

Input Attributes :

Value - attribute specifies an initial value for an input field

readonly - The input readonly attribute specifies that an input field is read-only, cannot be modified

disabled - A disabled input field is unusable and un-clickable, The value of a disabled input field will not be sent when submitting the form!

size - The input size attribute specifies the visible width, in characters, of an input field.

The default value for size is 20.

maxlength - The input maxlength attribute specifies the maximum number of characters allowed in an input field.

min and max - The input min and max attributes specify the minimum and maximum values for an input field.

multiple - The input multiple attribute specifies that the user is allowed to enter more than one value in an input field.

placeholder - The input placeholder attribute specifies a short hint that describes the expected value of an input field

required - The input required attribute specifies that an input field must be filled out before submitting the form.

step - The input step attribute specifies the legal number intervals for an input field.

autofocus - The input autofocus attribute specifies that an input field should automatically get focus when the page loads.

list - The input list attribute refers to a <datalist> element that contains pre-defined options for an <input> element.

autocomplete - The input autocomplete attribute specifies whether a form or an input field should have autocomplete on or off.

Multimedia

Multimedia tags in HTML are used to display media content like images, audio, and video on a webpage.

The `<audio>` tag is used to **play sound files on a webpage**.

- Controls, autoplay, loop, muted

The `<video>` tag is used to **play videos on a webpage**.

- Controls, autoplay, loop, muted

Source - Used to provide **multiple media formats**.

Favicon

A **Favicon (Favorite Icon)** is a small icon displayed in the **browser tab, bookmarks, and address bar** of a website.

<marquee> : The <marquee> tag in HTML creates a scrolling text or image effect within a webpage. It allows content to move horizontally or vertically across the screen, providing a simple way to add dynamic movement to elements.